

# Qiyuan (Vincent) Zhu

Tel: 2154303398 | Email: qiyuanzhu522@gmail.com | <https://www.linkedin.com/in/qiyuan-zhu-6171b9249/> | GitHub: <https://github.com/vincentvicente>

## EDUCATION

### Northeastern University

Master of Computer Science

Sep 2023-May 2026

### University of Pennsylvania

Master of Public Administration

Sep 2022-May 2023

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript/TypeScript, Java

**AI/LLM Engineering:** LLM SDKs - Anthropic, OpenAI, AWS Bedrock, Gemini, OpenRouter;

Agent Framework - LangGraphLangChain, FastAPI Nanobot, OpenClaw; Tool Use - Function Calling, ReAct, MCP;

Vector & RAG - pgvector, Pinecone, Chroma, FAISS, sentence-transformers;

Memory & Context Compression; Evaluation LangSmith, Ragas;

**Frameworks/Tools:** Backend - FastAPI, Node.js, Spring Boot; Micro Services;

API - REST/gRPC/WebSocket/SSE; Git - GitHub, Bitbucket; Docker; CI/CD - AWS CodeCommit, CodeBuild, ECR, CodeDeploy;

Message Queue - Kafka, RabbitMQ, AWS SNS/SQS;

Relational Database - MySQL, PostgreSQL(pgvector), AWS RDS; NoSQL - MongoDB, AWS DynamoDB;

Cache - Redis; Unit Test - pytest/Junit; Monitoring & Dashboard - Splunk

**Cloud Platform:** AWS(Bedrock, S3, EC2, RDS, DynamoDB, Lambda, Kinesis, SNS, SQS, RedShift, Glue, IAM, EMR,

CodeCommit, CodeBuild, ECR, CodeDeploy etc.), Azure, GCP

**AI Development Tools:** Claude Code, Cursor, ChatGPT, GitHub Copilot, MCP Inspector, LangGraph Studio

## PROFESSIONAL EXPERIENCES

### Vizzy Labs

Software Engineer Intern

Apr 2026-Present

- Developed a **multi-tenant LLM agent** fleet on a custom **FastAPI** Nanobot runtime with **Claude Opus 4.6**, packaging **12** production agents into a single **Docker** image differentiated by per-bot env vars, reducing image build/deploy cost by **80%**.
- Built a **7-step** durable Campaign **Auto-Pilot workflow** on **Hatchet**, featuring crash-recovery, server-side CEL event filtering, human-in-the-loop confirmation cards, and per-conversation concurrency control, cutting manual campaign-ops time by **60%** across **100+** active campaigns.
- Implemented a hierarchical **agent memory system** with automatic context compression at the **166k-token** threshold inside a **200k** window, sustaining coherent multi-day conversations across **5,000+** sessions.
- Designed a difficulty-aware **LLM router** across **Anthropic**, **Bedrock**, **OpenAI**, and **OpenRouter**, and exposed **20+** TikTok/Instagram analytics tools via a **FastAPI** and **FastMCP** server with less than **800 ms** average tool-call latency.

### Artisk Inc.

Software Engineer Intern

Oct 2025-Dec 2025

- Built a **Multi-Agent Orchestration System** using **Claude Sonnet 3.5**, implementing a **Dynamic Prompt Pipeline** that increased design intent recognition accuracy by **45%** through structured metadata extraction and Few-shot learning.
- Architected a high-concurrency **Asynchronous Task Queue** with **Meteor** and **MongoDB**, utilizing **WebSocket** for real-time status pushing; eliminated UI blocking for long-running AI operations and managed complex concurrency control.
- Integrated a **Multi-model Generative Pipeline**, implementing a **closed-loop feedback system** that captures *modelRevisedPrompt* to maintain "Visual Memory" across multi-turn design iterations.

## PROJECT HIGHLIGHTS

### AI Inbound Lead Pipeline

Feb 2026-Apr 2026

- Shipped an inbound-lead enrichment pipeline orchestrating **Claude Sonnet 4.6** / **Haiku 4.5** generation and **7 public APIs** with concurrent per-lead enrichment and per-fact provenance tracking, achieving **sub-30s** end-to-end latency at **~\$0.015** per lead.
- Engineered a **4-layer** hallucination-defense stack for grounded generation — per-fact confidence thresholding, post-generation entity & number verification with **auto-regeneration loops**, **multi-model fallback chain** and source-attributed UI rendering — verified by **129 unit/integration tests**.
- Built the surrounding **evaluation + HITL infrastructure**: **JSONL persistence** of every system/user prompt and model response for prompt regression, and a **human-in-the-loop dashboard** capturing edit-diffs and review latency to drive weekly prompt iteration.

### Video RAG Agent System

Jan 2026-Feb2026

- Architected an AI-powered **Video RAG agent** by building **3 MCP** servers with **12 custom tools** and **5 Claude AI skills**, enabling autonomous video processing, knowledge extraction, and natural-language Q&A over video content.
- Engineered an end-to-end data pipeline: video download, audio extraction (**FFmpeg**), speech-to-text transcription (**OpenAI Whisper**), semantic chunking (**NLP/TF-IDF**), and vector embedding (**OpenAI Embeddings**), transforming unstructured video into a searchable knowledge base.
- Implemented a **vector similarity search engine** with **cosine-similarity ranking** and **metadata filtering**, allowing users to semantically query across multi-video libraries and generate AI-driven topic summaries with source attribution.